

7 Fig. 7.1 shows an x-ray spectrum.

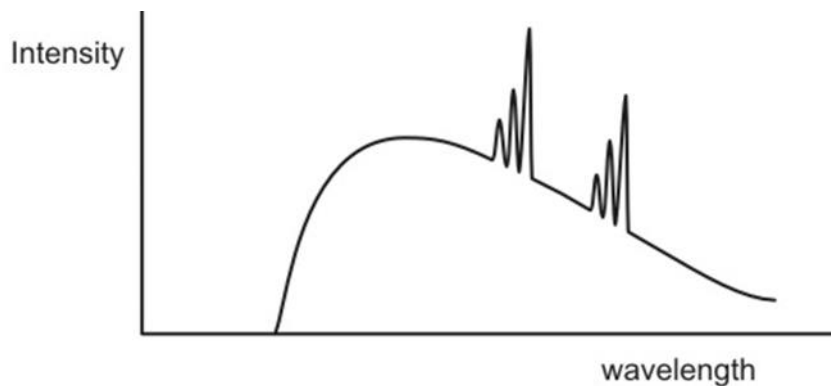


Fig. 7.1

Describe how the continuous spectrum is produced.

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.....[3]

[Total: 3]

		Solution: An electron with high speed/ KE interacts with the nucleus of a target metal and accelerates/decelerates. As electron is a charged particle, when it undergoes a rapid deceleration, it produces an x-ray photon of energy equal to the kinetic energy lost by the incident electron via Bremsstrahlung process. If an electron has excess kinetic energy after the deceleration it goes through a series of decelerations and produces photons of a (continuous) range of wavelengths.	1 1 1
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Section B

Answer **one** question from this section.

2	(a)	Fig. 2.1 shows a diagram of an electron gun.
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